EE2015: Electric Circuits and Networks

Tutorial 9 Answer Key

(October 24, 2025)

1. (a) $i_{\text{eff}} = \frac{4}{\sqrt{3}}$, (b) , (c)
2. (a) , (b)
3. $P_1 = -367.8\,\text{W}$, $P_2 = 160\,\text{W}$, $P_3 = P_4 = 0$, $P_5 = 207.8\,\text{W}$
4. 240 VA, 0.866 (leading), 25.98 Ω resistor in series with 212.2 μF
5. (a) 481.6 VA, (b) 462.4 W, (c) 134.6 VAR, (d) 0.96 (lagging)
6. 110.86 μF
7. $\mathbf{S} = 138.48 - j\,55.82\,\text{VA}$